Surname / *Van*: | Initials / *Voorletters*:

Student number / *Studentenommer*:

STICK BARCODE STICKER HERE

MEMO

FSK 116

Semester Test 1 - *Semestertoets 1*

Department of Physics, University of Pretoria
Departement Fisika, Universiteit van Pretoria

Date/*Datum*: 17/03/2017 **Time/*Tyd*: 2.5 hours/*ure*** **Total / *Totaal* : 75**

Examiner/*Eksaminator*: RQ Odendaal **Moderator/*Moderator*: Dr JM Nel**

$\checkmark \equiv 1$ mark $\quad \hat{\checkmark} \equiv \frac{1}{2}$ mark

---

Test register: Tear carefully on the line above and hand to the invigilator.
Toets register: Skeur versigtig op die lyn hierbo en handig dit in by die toesighouer.

FSK 116: Semester Test 1 / *Semestertoets 1* **17/03/2017**

Surname *Van*				Initials *Voorletters*			
Student number *Studentenommer*							
Venue *Lokaal*							
Signature *Handtekening*							

STICK BARCODE STICKER HERE

Please read the following carefully. ***Lees die volgende noukeurig***

1. Answer all the questions in the space provided on the question paper. *Antwoord alle vrae in die spasie gelaat op die vraestel.*

2. No pencil work will be marked. Do not use "tippex". *Geen potlood-werk sal gemerk word nie. Moet nie uitwis-vloeistof ("tippex") gebruik nie.*

3. If you need more space, please use the back of the pages, and clearly indicate where the question continues. *As u meer spasie benodig gebruik die agterkant van die bladsye. Dui duidelik aan waar die vraag aangaan.*

4. Where applicable you need to draw the necessary diagrams (e.g. free-body diagrams). *Teken diagramme waar nodig (bv. vryliggaams-diagramme).*

5. Your answer needs to be written in a systematic and logical fashion. You must show that you understand and can use the relevant physics. **Only substitute numerical values in the final step of calculations! A correct answer does not imply that you will receive full marks.** *U antwoorde moet sistematies en logies neergeskryf word. U moet bewys dat u die nodige fisika verstaan en kan toepas.* ***Numeriese waardes mag slegs in die laaste stap van 'n berekening vervang word! 'n Korrekte antwoord beteken nie dat u noodwendig vol punte sal kry nie.***

6. **Units must be used when numbers are substituted else marks will be subtracted.** ***Eenhede moet gebruik word wanneer numeriese waardes vervang word, andersins sal punte afgetrek word.***

7. The use of the cheat sheet is only allowed subject to the rules stated on it. *Die formuleblad mag slegs gebruik word onderhewig aan die reëls daarop gedruk.*

8. All exam regulations of the University of Pretoria apply, and any contravention of it will be treated in a serious manner. *Alle eksamen regulasies van die Universiteit van Pretoria geld en enige oortreding daarvan sal in 'n ernstige lig beskou word.*

Constants and formulas / ***Konstantes en formules***
$\varepsilon_0 = 8.854 \times 10^{-12}$ C^2/(N·m^2); $G = 6.674 \times 10^{-11}$ N·m^2/kg^2; $e = 1.602 \times 10^{-19}$ C; $k = 1.381 \times 10^{-23}$ J/K; $m_e = 9.109 \times 10^{-31}$ kg; $m_p = 1.673 \times 10^{-27}$ kg; $R = 8.314$ J/(mol·K); $N_A = 6.02 \times 10^{23}$ /mol; 1 atm = 101.3 kPa; 1 u = $1.6605677 \times 10^{-27}$ kg; $c = 3 \times 10^8$ m/s; $V_{\text{sphere/sfeer}} = \frac{4}{3}\pi r^3$; $F_{gravity} = \frac{Gm_1m_2}{r^2}$; 1 litre = 1000 cm^3; $M_{\text{Earth}} = M_{\text{Aarde}} = 5.97 \times 10^{24}$ kg, ρ(water) = 1 g/cm^3; $r_{\text{Earth}} = r_{\text{Aarde}} = 6.37 \times 10^6$ m

---

1 **Question 1 /** ***Vraag 1*** [6]

In quantum physics a jiffy is, as defined by Edward Harrison, the time it takes for light to travel one femtometer in vacuum, which is approximately the size of a nucleon. The speed of light is given as 299 792 458 m/s in vacuum.
In kwantummeganika is 'n "jiffy", soos gedefinieer deur Edward Harrison, die tyd wat lig neem om een femtometer in vakuum te beweeg, wat ongeveer die grootte van 'n nukleon is. Die spoed van lig is 299 792 458 m/s in vakuum.

(a) Express a jiffy in seconds to two decimal places. All conversion factors must be shown in your calculation.
Druk 'n "jiffy" uit in sekondes tot twee desimale plekke. Alle omskakelingsfaktore moet getoon word in jou berekening. (2)

1 fm = 10^{-15} m

Time is distance travelled divided by speed

$t = \frac{d}{c} = \frac{10^{-15}\text{ m}}{299\ 792\ 458\text{ m/s}} = 3.34 \times 10^{-24}$ s = 1 jiffy

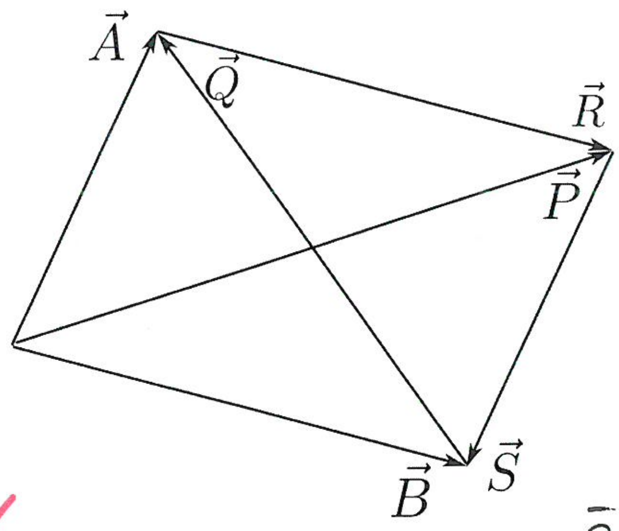
(b) Express one month of 31 days in the unit of a jiffy. All conversion factors must be shown in your calculation.
Druk een maand met 31 dae uit in die eenheid van 'n "jiffy". Alle omskakelingsfaktore moet getoon word in jou berekening. (4)

$$1\,\text{month} \times \frac{31\,\text{days}}{1\,\text{month}} \times \frac{24\,\text{h}}{1\,\text{day}} \times \frac{60\,\text{min}}{1\,\text{h}} \times \frac{60\,\text{s}}{1\,\text{min}} \times \frac{1\,\text{jiffy}}{3.34 \times 10^{-24}\,\text{s}}$$

$$= 8.02 \times 10^{29}\ \text{jiffies}$$

Question 2 / *Vraag 2* [5]

Refer to the figure below. Express the vectors $\vec{P}$, $\vec{R}$, $\vec{S}$, and $\vec{Q}$ in terms of vectors $\vec{A}$ and $\vec{B}$.
Verwys na die figuur hieronder. Skryf die vektore $\vec{P}$, $\vec{R}$, $\vec{S}$, en $\vec{Q}$ in terme van vektore $\vec{A}$ en $\vec{B}$.



$\bar{R} = \bar{B}$

$\bar{S} = -\bar{A}$

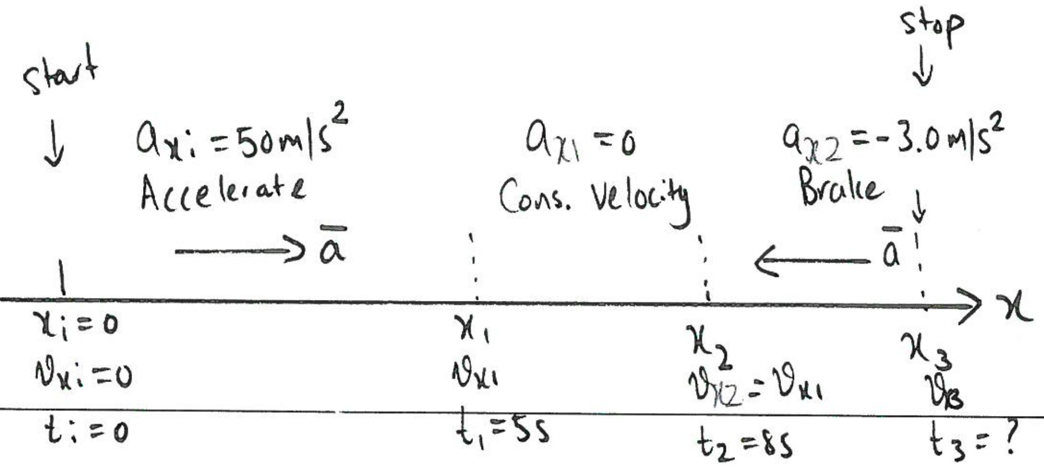
$\bar{P} = \bar{A} + \bar{R} = \bar{A} + \bar{B}$

$\bar{Q} = \bar{A} - \bar{B}$

Question 3 / *Vraag 3* [10]

A rocket sled accelerates from rest at 50 m/s^2 for 5.0 s. It then moves at constant velocity for another 3.0 s, then it deploys a braking parachute and slows down with an acceleration of 3.0 m/s^2 opposite in direction to that of the motion, until it comes to a stop. What is the maximum velocity of the rocket sled and the total distance it travelled?

'n Vuurpyl-aangedrewe slee vernel vanuit rus teen 50 m/s^2 vir 5.0 s. Dit beweeg dan teen 'n konstante snelheid vir 3.0 s, waarna 'n rem-valskerm uitgeskiet word wat veroorsaak dat die slee al stadiger beweeg teen 'n versnelling van 3.0 m/s^2 teenoorgesteld in rigting van dié van beweging totdat dit tot stilstand kom. Wat is die maksimum snelheid van die slee, sowel as die totale afstand wat dit beweeg het?



Max. velocity is after the acceleration; that is v_{x1}

$v_{x1} = v_{xi} + a_{xi} t_1 = 0 + (50 m/s^2)(5s) = 250 m/s = v_{max}$

To find the total distance we will have to find the position at t_1, t_2 and t_3, i.e. x_1, x_2 and x_3

$x_1 = x_i + v_{xi}t_1 + \frac{1}{2}a_{xi}t_1 = 0 + 0 + \frac{1}{2}(50\,m/s^2)(5s)^2 = 625\,m$

When moving at constant velocity, $v_{xi} = 250\,m/s$

$x_2 = x_1 + v_{xi}(t_2 - t_1)$

$= 625\,m + (250\,m/s)(3s)$

$x_2 = 1375\,m$

We do not know how long the braking lasts.

We do know that $v_{x3} = 0$ and $v_{x2} = v_{x1}$

$v_{x3}^2 = v_{x2}^2 + 2a_{x2}(x_3 - x_2)$

$x_3 = x_2 + \dfrac{v_{x3}^2 - v_{x2}^2}{2a_{x2}}$

$= 1375\,m + \dfrac{0 - (250\,m/s)^2}{2(-3.0\,m/s^2)}$

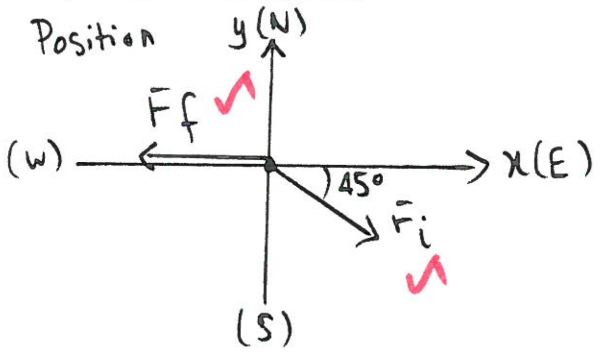
$x_3 = 11\,792\,m$

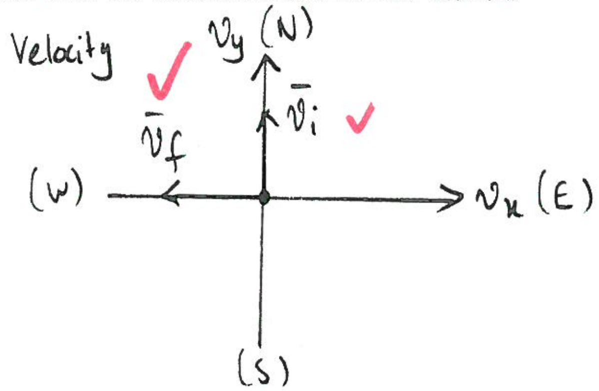
x_3 is the total distance travelled.

Question 4 / ***Vraag 4*** [11]

A runner starts at a position of 1 km east and 1 km south of the town-hall. Her initial velocity is 1.5 m/s north. After 35 minutes, she is at a position 2 km west of the town-hall and her velocity at this moment is 2.1 m/s west.

'n Atleet begin by 'n posisie van 1 km oos en 1 km suid van die stadsaal. Haar aanvangsnelheid is 1.5 m/s noord. Na afloop van 35 minute, is sy by 'n posisie 2 km wes van die stadsaal en haar snelheid is op dié oomblik 2.1 m/s wes.





(a) Calculate her **average** velocity for the 35 minute run, using the town-hall as the origin of your coordinate system.
*Bereken haar **gemiddelde** snelheid vir die 35 minute roete, en gebruik die stadsaal as die oorsprong van jou koördinaat-stelsel.* (6)

$\bar{r}_i = (1\,km)\hat{\imath} - (1\,km)\hat{\jmath}$; $\bar{r}_f = -(2\,km)\hat{\imath}$

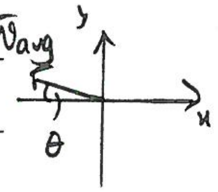
Displacement : $\Delta \bar{r} = \bar{r}_f - \bar{r}_i = -(2\text{km})\hat{i} - [(1\text{km})\hat{i} - (1\text{km})\hat{j}]$

$= -(3\text{km})\hat{i} + (1\text{km})\hat{j}$

$\bar{v}_{avg} = \frac{\Delta \bar{r}}{\Delta t} = -\left(\frac{3\text{km}}{35\text{min}} \times \frac{60\text{min}}{1\text{h}}\right)\hat{i} + \left(\frac{1\text{km}}{35\text{min}} \times \frac{60\text{min}}{1\text{h}}\right)\hat{j}$

$= -(5.1\text{ km/h})\hat{i} + (1.7\text{ km/h})\hat{j}$

$|\bar{v}_{avg}| = 5.37\text{ km/h} \qquad \theta = 18^\circ$ above $-x$-axis



(b) What is the **average** acceleration for the 35 minute run using the same coordinate system as in (a)?
Wat is die ***gemiddelde*** *versnelling vir die 35 minute roete as jy dieselfde koördinaat-stelsel gebruik as in (a)?* (5)

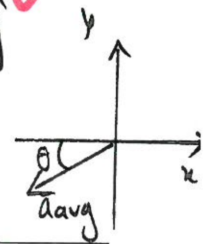
$\Delta \bar{v} = \bar{v}_f - \bar{v}_i$

$\Delta \bar{v} = -(2.1\text{m/s})\hat{i} - (1.5\text{m/s})\hat{j}$

$\bar{a}_{avg} = \frac{\Delta \bar{v}}{\Delta t} = -\left[\frac{2.1\text{m/s}}{35\text{min}} \times \frac{1\text{min}}{60\text{s}}\right]\hat{i} - \left[\frac{1.5\text{m/s}}{35\text{min}} \times \frac{1\text{min}}{60\text{s}}\right]$

$= -(1.0 \times 10^{-3}\text{ m/s}^2)\hat{i} - (7.1 \times 10^{-4}\text{ m/s}^2)\hat{j}$

$|\bar{a}_{avg}| = 1.2 \times 10^{-3}\text{ m/s}^2 \qquad \theta = 35.4^\circ$ below $-x$-axis



5 **Question 5 / *Vraag 5*** [4]

An object moves in one dimension, say in the x-direction, with a **constant acceleration** a_x. At time $t_i = 0$ the object is at a position x_i and is travelling with a velocity of v_{xi}. At a later time t, the object is at a position x, and moving with velocity v_x. Given that the displacement is given as $\Delta x = v_{xi}t + \frac{1}{2}a_x t^2$ and the velocity is given by $v_x = v_{xi} + a_x t$, prove that $v_x^2 = v_{xi}^2 + 2a_x(x - x_i)$.
'n Voorwerp beweeg in een dimensie, veronderstel die x-rigting, teen 'n ***konstante versnelling*** *a_x. By tyd $t_i = 0$ is die voorwerp by posisie x_i en beweeg met 'n snelheid v_{xi}. Op 'n latere tydstip t, vind ons die voorwerp by posisie x, en dit beweeg met 'n snelheid v_x. Gegee dat die verplasing bereken kan word met $\Delta x = v_{xi}t + \frac{1}{2}a_x t^2$ en die snelheid met $v_x = v_{xi} + a_x t$, bewys dat $v_x^2 = v_{xi}^2 + 2a_x(x - x_i)$.*

$v_x = v_{xi} + a_x t \quad - (1) \qquad x - x_i = v_{xi}t + \frac{1}{2}a_x t^2 \quad - (2)$

From (1) : $t = \frac{v_x - v_{xi}}{a_x} \quad - (3) \qquad \Delta x = x - x_i$

Substitute (3) into (1) : $x - x_i = v_{xi}\left(\frac{v_x - v_{xi}}{a_x}\right) + \frac{1}{2}a_x\left(\frac{v_x - v_{xi}}{a_x}\right)^2$

$x - x_i = \frac{v_{xi}v_x - v_{xi}^2}{a_x} + \frac{1}{2}\frac{a_x}{a_x^2}(v_x - v_{xi})^2$

$a_x(x - x_i) = v_{xi}v_x - v_{xi}^2 + \frac{1}{2}[v_x^2 - 2v_{xi}v_x + v_{xi}^2]$

$$2a_x(x-x_i) = \cancel{2v_{xi}v_x} - 2v_{xi}^2 + v_x^2 - \cancel{2v_{xi}v_x} + v_{xi}^2$$

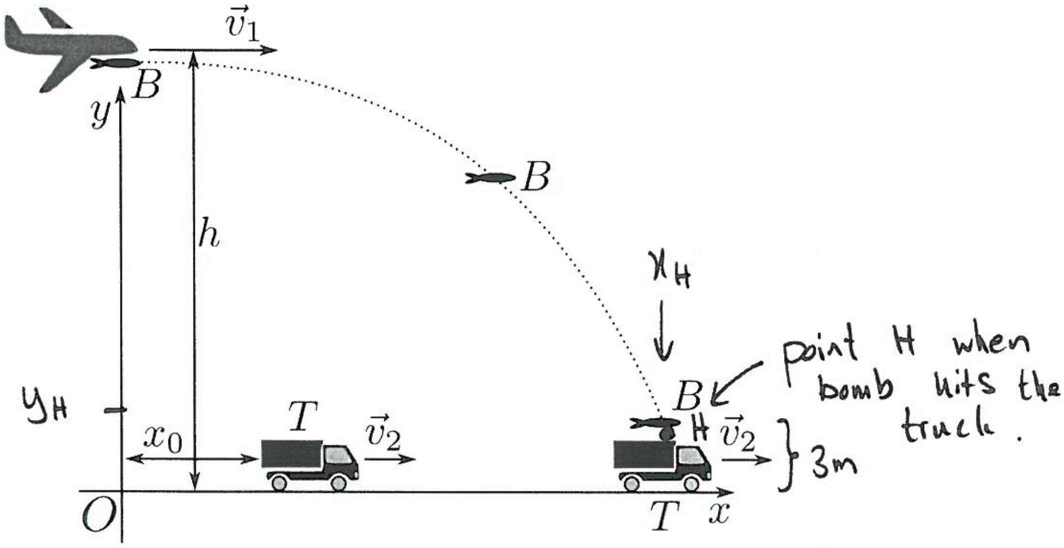
$$-v_x^2 = -v_{xi}^2 - 2a_x(x-x_i)$$

$$v_x^2 = v_{xi}^2 + 2a_x(x-x_i)$$

6 **Question 6 / *Vraag 6*** [6]

A bomber (see figure) is flying level at a constant speed $v_1 = 72$ m/s, at a height of $h = 103$ m. When directly over the origin O, a bomb B is released and strikes the truck T, which is moving along a level road (x-axis) with constant speed. At the the instant the bomb is released the truck is a distance $x_0 = 125$ m from O. Find the value of v_2 and the time of flight of bomb B. Assume that the truck is 3 m high, and there is no air resistance.

'n Bomwerper (sien figuur) vlieg op 'n gelyke hoogte van $h = 103$ m, teen 'n konstante spoed $v_1 = 72$ m/s. Wanneer dit direk oor die oorsprong O is, word 'n bom B vrygelaat en dit tref 'n trok T, wat langs 'n gelyke pad (die x-as) beweeg het teen 'n konstante spoed. Op die oomblik toe die bom vrygelaat is, is die trok 'n afstand $x_0 = 125$ m vanaf O. Vind die waarde van v_2 en die vlugtyd van die bom B. Neem aan die trok het 'n hoogte van 3 m, en daar is geen lugweerstand nie.



Use the coordinate system as chosen in the sketch.

For the bomb; it hits at point H when $y_H = 3$m

but y_i for the bomb is $y_i = 103$ m.

$$y_H = y_i + v_{yi}t_H + \frac{1}{2}a_y t_H^2 \quad ; \text{ but } v_{yi} = 0, \; a_y = -g$$

$$\Rightarrow \quad y_H = y_i + 0 - \frac{1}{2}g t_H^2$$

$$t_H^2 = \frac{2(y_i - y_H)}{g}$$

$$t_H = \sqrt{\frac{2(y_i - y_H)}{g}} = \sqrt{\frac{2(103\,m - 3\,m)}{9.8\,m/s^2}}$$

$$t_H = 4.52\,s$$

For the bomb $x_H = v_{xi} t_H + \frac{1}{2} a_x t_H$ but $a_x = 0$

$$x_H = v_{xi} t_H$$

$$x_H = v_1 t_H = (72\,m/s)(4.52\,s) = 325.4\,m$$

The truck has the same x-position as the bomb at point H:

$x_H = v_2 t_H + x_{i,T}$ (constant velocity for the truck)

$$\Rightarrow v_2 = \frac{x_H - x_{i,T}}{t_H} \qquad x_{i,T} = x_0$$

$$= \frac{x_H - x_0}{T_H}$$

initial position for truck.

$$= \frac{325.4\,m - 125\,m}{4.25\,s}$$

$$v_2 = 44.3\,m/s$$

7 **Question 7 / *Vraag 7*** [9]

A particle moving in the xy-plane has x and y components of velocity given by

$$v_x = b_1 + c_1 t \quad \text{and} \quad v_y = b_2 + c_2 t,$$

where x and y are measured in meters and t in seconds.

'n Deeltjie beweeg in die xy-vlak en het x en y komponente vir die snelheid gegee deur

$$v_x = b_1 + c_1 t \quad \text{en} \quad v_y = b_2 + c_2 t,$$

waar x en y in meters gemeet word en t in sekondes gemeet word.

(a) What are the dimensions and units of the constants b_1, b_2, c_1 and c_2?
Wat is die dimensies en eenhede van die konstantes b_1, b_2, c_1 en c_2? (2)

b_1 and b_2 have dimensions [Velocity] = $[\frac{L}{T}]$

unit of m/s

c_1 and c_2 have dimensions [Acceleration] = $[\frac{L}{T^2}]$

unit of m/s^2

(b) Find functions $x(t)$ and $y(t)$ (positions as functions of time).
Vind funksies $x(t)$ en $y(t)$ (posisies as funksies van tyd). (3)

$\frac{dx}{dt} = b_1 + c_1 t$

$\Rightarrow \int_{x_i}^{x} dx' = \int_0^t (b_1 + c_1 t')\, dt'$

$[x']_{x_i}^{x} = b_1 \int_0^t 1\, dt' + c_1 \int_0^t t'\, dt'$

$x - x_i = b_1 [t']_0^t + c_1 [\tfrac{1}{2} t'^2]_0^t$

$x = x_i + b_1 t + \tfrac{1}{2} c_1 t^2$

and similar for $y = y_i + b_2 t + \tfrac{1}{2} c_2 t^2$

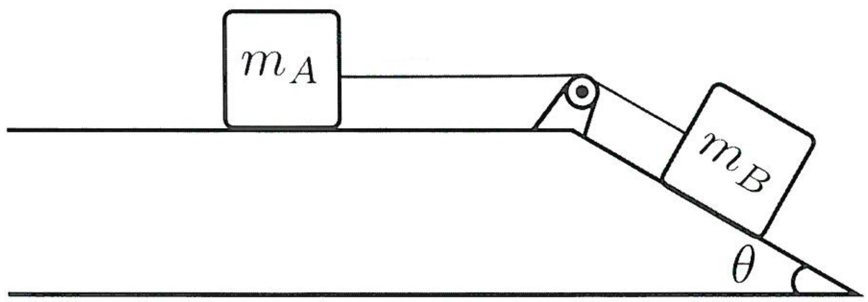
(c) Denoting the total acceleration as $\vec{a}$, find expressions for the magnitude of and direction (angle) of $\vec{a}$.
Indien die totale versnelling geskryf kan word as $\vec{a}$, vind uitdrukkings vir die grootte en rigting (hoek) van $\vec{a}$. (4)

$a = \left[\left(\frac{d^2x}{dt^2}\right)^2 + \left(\frac{d^2y}{dt^2}\right)^2\right]^{1/2} = (c_1^2 + c_2^2)^{1/2} = \sqrt{c_1^2 + c_2^2}$

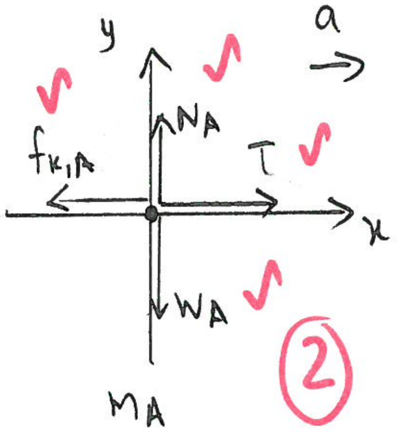
$\tan\alpha = \frac{\left|\frac{d^2y}{dt^2}\right|}{\left|\frac{d^2x}{dt^2}\right|} = \frac{c_2}{c_1}$ where α is the angle $\bar{a}$ makes with the pos. x-axis

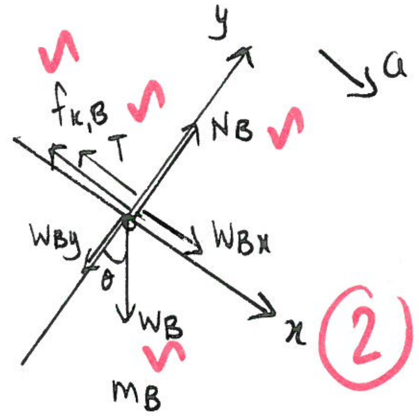
8 Question 8 / *Vraag 8* [12]

The figure shows an arrangement with two identical crates, m_A and m_B, with mass 40 kg each. Both objects have the same coefficients of friction given as $\mu_k = 0.15$ and $\mu_s = 0.25$. If the system is allowed to move from rest, find the acceleration of the boxes and the tension in the massless cord. The angle $\theta = 30°$, and the pully is ideal (frictionless).
You must use the component method together with the appropriate Newton law and free-body diagrams to solve the problem.
Die figuur toon 'n opstelling met twee identiese kratte, m_A en m_B, met massa 40 kg elk. Beide voorwerpe het koëffisiënte van wrywing gegee deur $\mu_k = 0.15$ and $\mu_s = 0.25$. Indien die stelsel vrygelaat word vanuit rus, bereken die versnelling van die kratte sowel as die spankrag in die massalose koord wat hulle verbind. Die hoek $\theta = 30°$, en die katrol is ideaal (wrywingloos).
Jy moet die komponent-metode gebruik tesame met die nodige Newton wet en vry-liggaam diagramme om die probleem op te los.



Common acceleration a





(12)

For m_A: y-components $\Sigma F_y = 0$

$N_A - W_A = 0 \Rightarrow N_A = W_A = m_A g$

x-components $\Sigma F_x = m_A a$

$T - f_{k,A} = m_A a$

$T - \mu_k N_A = m_A a$

$T - \mu_k m_A g = m_A a$ — ①

For m_B: y-components $\Sigma F_y = 0$

$N_B - W_{By} = 0$

$N_B = W_{By} = m_B g \cos\theta$

x-components $\Sigma F_x = m_B a$

$W_{Bx} - T - f_{k,B} = m_B a$

$m_B g \sin\theta - T - \mu_k N_B = m_B a$

$m_B g \sin\theta - T - \mu_k m_B g \cos\theta = m_B a$ —②

①+②: $m_B g \sin\theta - \mu_k m_A g - \mu_k m_B g \cos\theta = (m_A + m_B) a$

$$a = \frac{m_B g \sin\theta - \mu_k m_A g - \mu_k m_B g \cos\theta}{m_A + m_B}$$

$$= \frac{(40kg)(9.8m/s^2)\sin 30^\circ - (0.15)(9.8m/s^2)[40kg + (40kg)(\cos 30^\circ]}{80kg}$$

$a = 1.08 m/s^2$

From ①: $T = m_A(a + \mu_k g) = (40kg)[1.08m/s^2 + (0.15)(9.8m/s^2)]$

$T = 102 N$

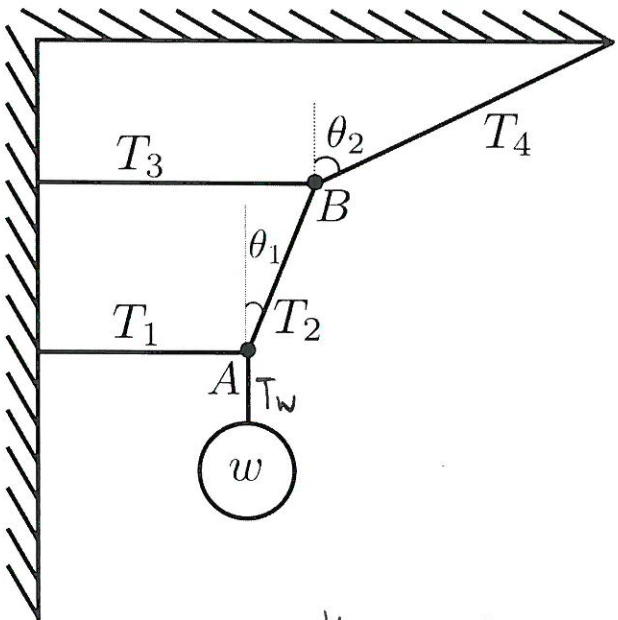
9 Question 9 / *Vraag 9* [12]

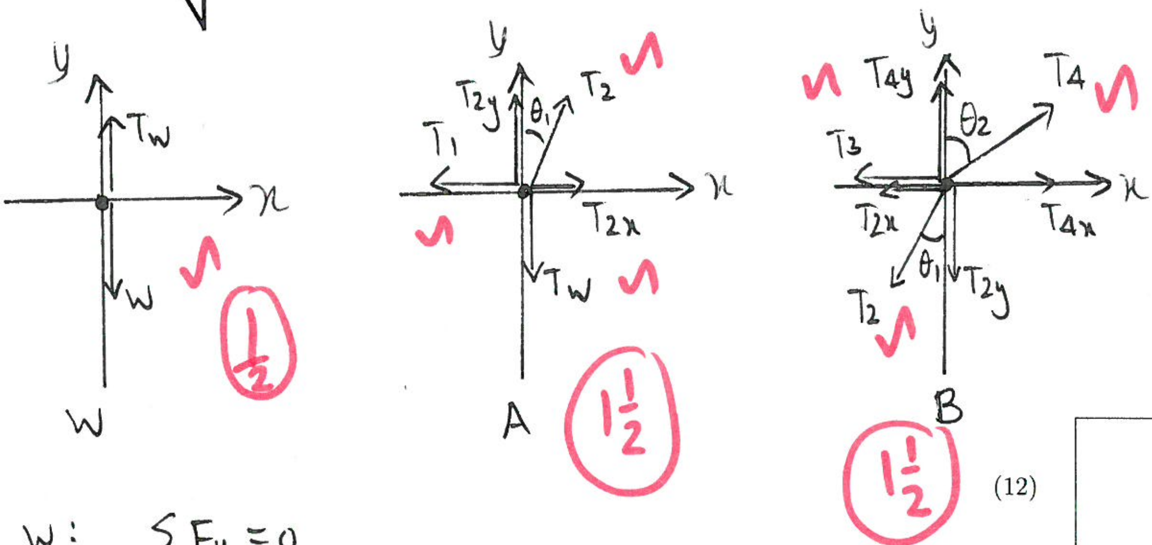
The figure shows an arrangement of (massless) strings attached to a singular mass of weight $w = 80$ N. The whole arrangement is in equilibrium. Find T_1, T_2, T_3 and T_4. The angles are $\theta_1 = 25°$ and $\theta_2 = 55°$.

Use the conventional coordinate system of horizontally to the right to be the positive x-direction, and vertically upwards to be the positive y-direction. **You must use the component method together with the appropriate Newton law and free-body diagrams (for points A and B and the mass w) to solve the problem.**

Die figuur toon 'n opstelling van (massalose) toutjies en 'n enkele massa met gewig $w = 80$ N. Die hele opstelling is in ewewig. Vind T_1, T_2, T_3 en T_4. Die hoeke in die opstelling is $\theta_1 = 25°$ en $\theta_2 = 55°$.

Gebruik die konvensionele koördinaat-stelsel met na regs en horisontaal die positiewe x-rigting, en vertikaal-opwaarts die positiewe y-rigting. ***Jy moet die komponent-metode gebruik tesame met die nodige Newton wet en vry-liggaam diagramme (vir punte A end B en die massa w) om die probleem op te los.***





(12)

For w: $\Sigma F_y = 0$

$T_w = W = 80N$ ✓

Point A: y-components: $\sum F_y = 0$

$$T_{2y} - T_W = 0$$
$$T_2 \cos\theta_1 = T_W$$
$$T_2 = \frac{T_W}{\cos\theta_1} = \frac{80N}{\cos 25^\circ} = 88.3N$$

x-components: $\sum F_x = 0$

$$T_{2x} - T_1 = 0$$
$$T_2 \sin\theta_1 = T_1$$
$$T_1 = (88.3N)\sin 25^\circ = 37.3N$$

(37.4N)

Point B: y-components $\sum F_y = 0$

$$T_{4y} - T_{2y} = 0$$
$$T_4 \cos\theta_2 = T_2 \cos\theta_1$$
$$T_4 = \frac{T_2 \cos\theta_1}{\cos\theta_2}$$
$$T_4 = (88.3N)\frac{\cos 25^\circ}{\cos 55^\circ}$$
$$T_4 = 139.5\,N \quad (140N)$$

x-components $\sum F_x = 0$

$$T_{4x} - T_{2x} - T_3 = 0$$
$$T_3 = T_{4x} - T_{2x}$$
$$= T_4 \sin\theta_2 - T_2 \sin\theta_1$$
$$= (139.5N)\sin 55^\circ - (88.3N)\sin 25^\circ$$
$$T_3 = 76.9N \quad (77N)$$

END OF TEST / EINDE VAN TOETS

